

Group 5 - Image Dependent Annotation Variability

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Abstract. The analysis and proper segmentation of medical images are crucial to both diagnosis and treatment of patients. As segmentation is usually performed manually by human experts, significant variability often arises as a result. This project aims to investigate the feasibility of a deep learning model in predicting the degree of disagreement among researchers given a dataset consisting of various medical images. Each image is annotated by 2-6 medical segmentation experts and labeled with a disagreement score. This score is subsequently used to train a convolutional neural network to predict the disagreement scores of non-annotated retinal images. Through the examination and evaluation of these results, we gain insight into the effectiveness of models measuring annotation uncertainty in medical image analysis.

Keywords: Image Annotation · Disagreement Score · Convolutional Neural Network · Regression Model

1 Introduction

1.1 Application and Problem Description

Annotation disagreement, also known as the inconsistent labeling of an image, is a common phenomenon in medical image analysis. As images are often manually labeled by medical experts, differences in the level of experience and interpretation along with the general quality of the image may lead to disagreements. Diagnostic models heavily depend on the "ground truth" of an image, thus, variability among ground-truth annotations can lead to heavy inconsistencies in the model. Machine learning models trained on such data may have biases, generalizations, and even produce inaccurate or unreliable predictions due to disagreement. As a result, addressing disagreement is essential for improving model effectiveness, ensuring proper clinical decisions for medical applications.

1.2 Motivation

Current high-performance medical machine learning models rely heavily on high-quality datasets for training to produce accurate and reliable outputs. Due to

strict quality requirements of the datasets, along with the ever-increasing need for medical models, obtaining annotated datasets of this caliber can be a difficult due to the time and cost needed for highly experienced annotators. As a result, datasets that exhibit high annotation disagreement are common, and models trained on such data may end up being incorrect and unreliable. Through analyzing and predicting image annotation disagreement, datasets can be subsequently quantified and filtered, allowing models to be trained on subpar datasets while maintaining correctness and reliability at a reasonable cost [1].

1.3 Background and Related Work

Medical Image Segmentation and the Role of Convolutional Neural Networks (CNNs) Medical image segmentation, also known as the "identification of organ or lesion pixels from medical images", is one of the most important and complex steps in medical imaging [2]. Images produced by CT or MRI scans require extensive feature extraction to obtain useful information for disease diagnosis and treatment [2]. As a result, segmentation is commonly performed manually by medical professionals; however, due to the complexity of the process and the subjective nature of images, this is an arduous task. Due to the issues mentioned above, CNN-based models have been introduced into medical imaging, resulting in exceptional performance; however, the challenge of poor-quality datasets remain a hurdle for CNNs.

Disagreement in Fundus Images The fundus is a part of the eyeball that contains vital structures such as the retina and optic nerve, and thus plays a critical role in eye health. As a result, many serious eye diseases such as glaucoma and macular degeneration can be detected early through medical imaging of the fundus [3]. However, detecting such diseases is a difficult, time-consuming, and subjective problem, resulting in a high degree of annotation disagreement between experts.

1.4 Overview of the Report

The rest of the report is divided as follows. Firstly, the different datasets used in this report will be discussed and explored in **Section 2**. Secondly, the proposed methods and algorithmic details will be covered in **Section 3**. Quantitative and qualitative results will be displayed in **Section 4**, alongside the accomplishments of the report in **Section 5**. Conclusions and further discussions of the report will be explored in **Section 7, 8** respectively.

2 Materials

Two data sets were used to train and test our model: Retinal Fundus Images for Glaucoma Analysis (RIGA) data set[4], and Retinal Vessel Reference

(CHASE) data set[5]. (See appendix 1 and appendix 2 for more information.) RIGA contains 195 original images of the retina. Each image had its optic cup and disc boundaries manually annotated by six ophthalmologists, resulting in $195 + 6(195) = 1365$ total images. To ensure consistency in their annotations, the standard deviation was calculated for each image and outliers were discarded. CHASE is a data set consisting of retinal images of 14 children, resulting in a dataset of 28 original retinal images (2 for each child). Each image is accompanied by manual vessel segmentations performed by two human observers, resulting in a total of $28 + 2(28) = 84$ images. These annotations mark the boundaries of blood vessels and shows blood vessels in white and everything else in black.

3 Methods

The method used in this report is to use a CNN model to predict disagreement values, more specifically Pytorch’s resnet18 model [6] [7]. It is necessary to have cleaned and annotated data in order for the model to be trained to predict disagreement values. Both RIGA and CHASE were previously cleaned and annotated; therefore, both are sufficient for training. Once data is acquired, disagreement values can be subsequently calculated.

To calculate the disagreement values, annotated images are first converted to a grayscale image which is then converted to binary.

Algorithm 1 Image To Binary Algorithm

```

1: if  $i$  is an image then
2:    $i \leftarrow \text{grayscale}$ 
3:    $i \leftarrow \text{binary}$ 
4: end if

```

The binary images would then be compared to other binary images and the difference between two images would be their disagreement value. To get the disagreement value for the image, every annotation of an image would be compared to all other annotations of the same image. The average of these disagreement values is then considered the disagreement value of the image.

Algorithm 2 Disagreement Algorithm

```

1: if  $i$  is an annotation of  $a$  and  $j$  is an annotation of  $a$  then
2:    $\text{difference} \leftarrow i - j$ 
3:    $\text{annotations} \leftarrow \text{annotations} + 1$ 
4: end if
5:  $\text{disagreement} \leftarrow \text{difference} / \text{annotations}$ 

```

The disagreement values and their corresponding images are then split into training and validation data sets. Approximately 80% of the images were used as training data, and the remaining 20% was used to validate the model. The code for training our model was adapted from PyTorch [8].

The model is trained in 75 epochs. During each epoch, the model is exposed to 16 images from the training set and updates its parameters accordingly. In the same epoch, the model is then given 16 images from the validation set to predict. The error (loss) in the prediction serves as our benchmark in how the model is doing. After each epoch, the loss is expected to decrease as the model better understands the data.

Algorithm 3 Model Training Algorithm

```

1:  $trainSet \leftarrow trainingimages$  ▷ Split into batches of 16 images
2:  $validationSet \leftarrow validationimages$  ▷ Split into batches of 16 images
3:  $model$  ▷ CNN model
4: while  $epoch$  less than 75 do
5:    $model \leftarrow trainSet$ 
6:    $model \leftarrow updatedParameters$ 
7:    $model \leftarrow validationSet$ 
8:    $print \leftarrow modelLoss$ 
9:    $epoch \leftarrow epoch + 1$ 
10: end while

```

4 Results

Once the model is trained, the model can now predict, to some degree, the level of disagreement annotators may have when annotating an image from the RIGA dataset, shown in Figure 1. The correlation coefficient (r) for our model was approximately 0.81818, the mean absolute error (MAE) approximately 0.008137, and the mean square error (MSE) approximately 0.000129. To further quantify the MAE , Figures 2 and 3 shows the 5 best and worst predictions in terms of MAE . It should be noted that a model was also trained on CHASE; however, the lack of data made it difficult to create a sensical model (as seen in 4) and no further investigation was carried out for this dataset.

It should also be noted that while testing across different devices, there tends to be some variance for these results as the random seed used for training may vary across devices. Thus, if a device generates a "good" seed, satisfactory results may be produced despite having a poor model. To account for this, dropout and batch normalization layers were added, and the number of epochs was adjusted. With the seed, both 25 and 75 epochs worked best; however, results were more stable when training without a seed at 75 epochs.

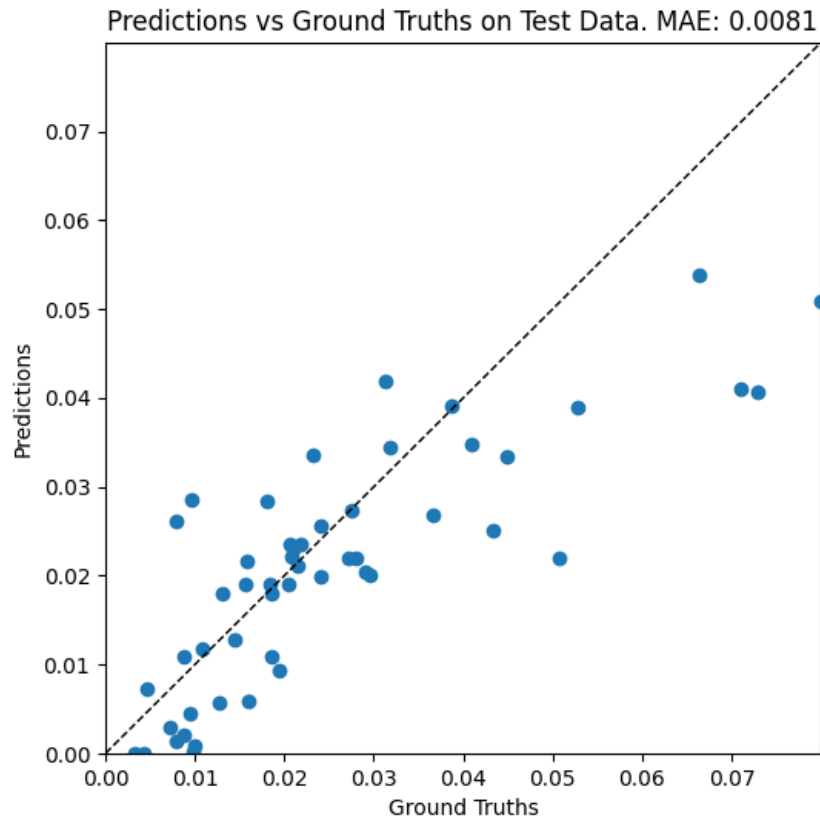


Fig. 1. The model's predictions plotted against ground truths for RIGA. The overall Mean Absolute Error (MAE) of the model is 0.008137

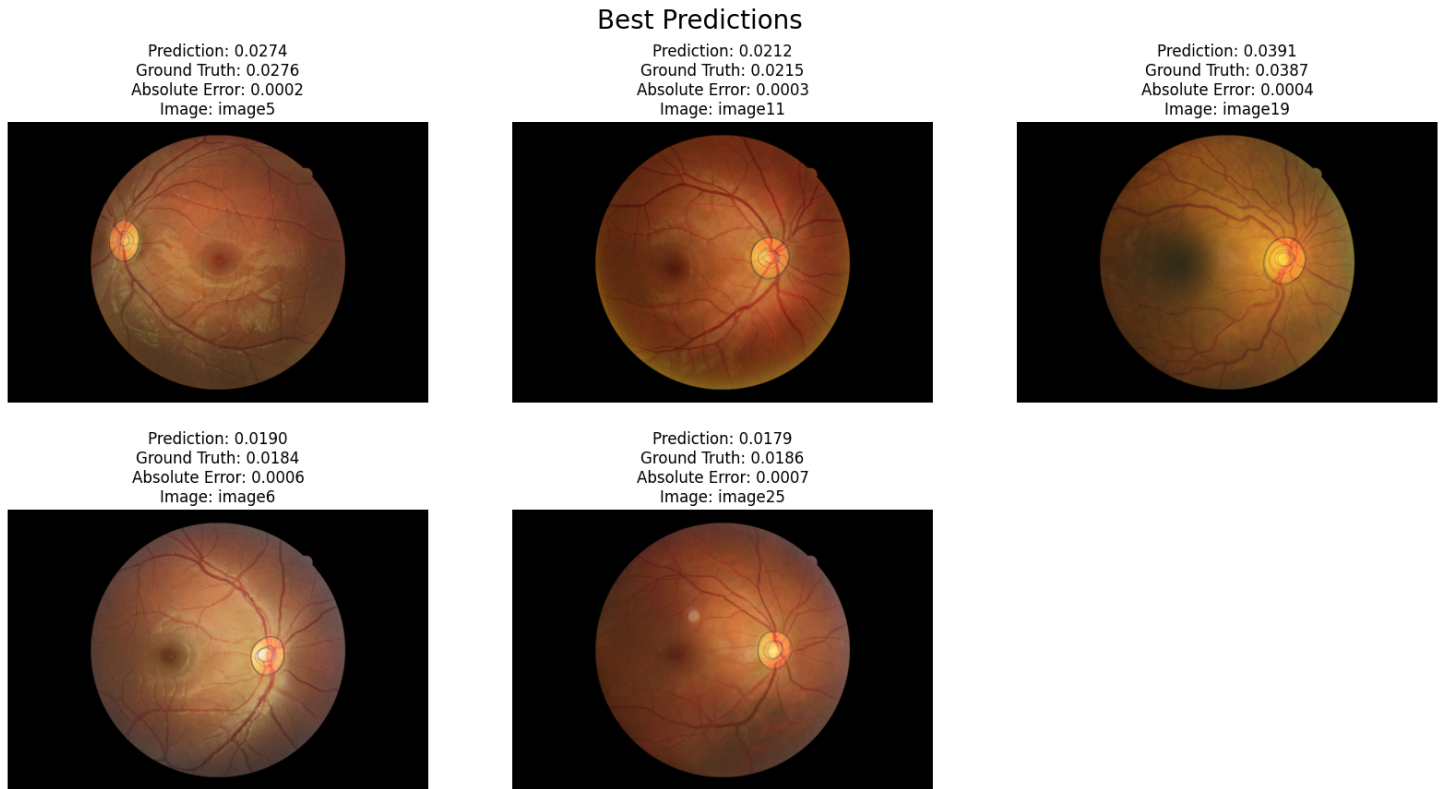


Fig. 2. The 5 best predictions from best to worst. Annotations are faintly overlaid to give a frame of reference.

Worst Predictions

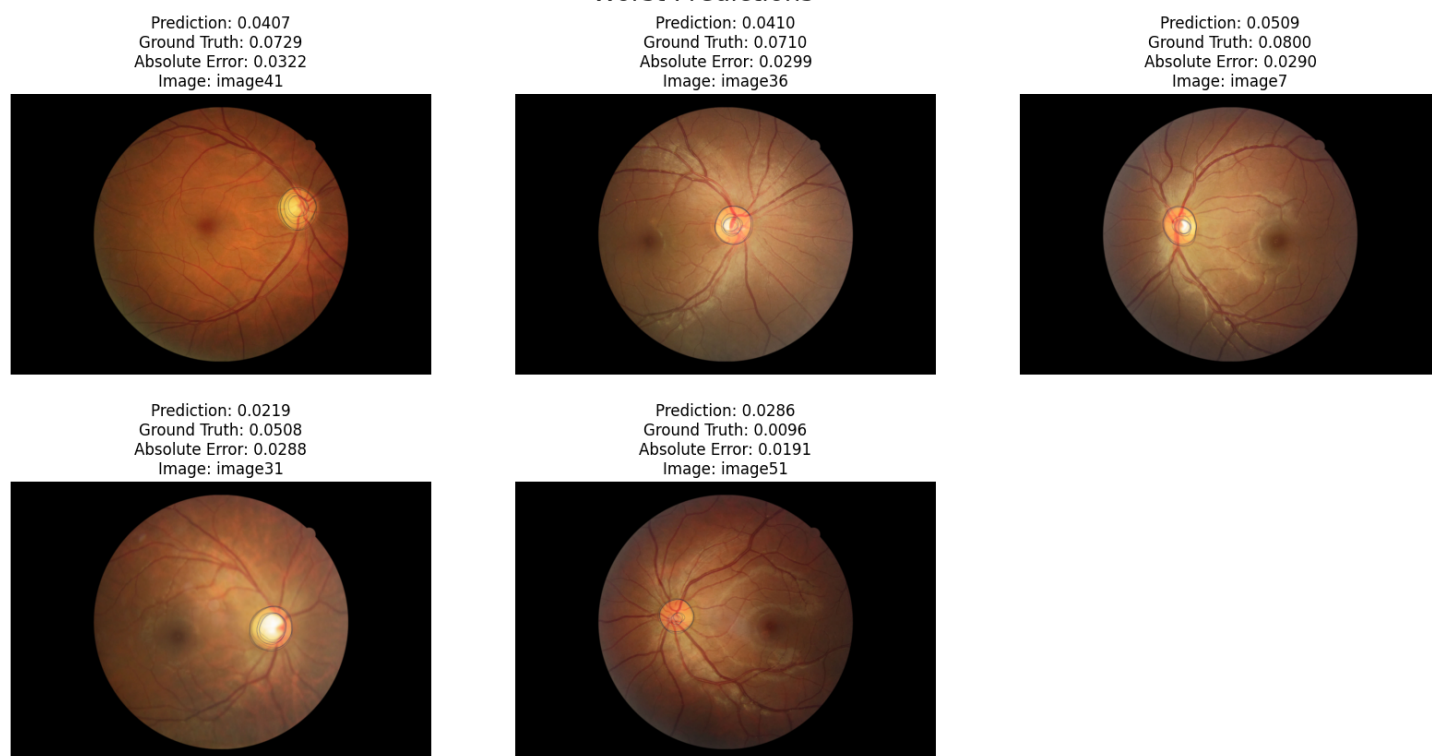


Fig. 3. The 5 worst predictions from worst to best. Annotations are faintly overlaid to give a frame of reference.

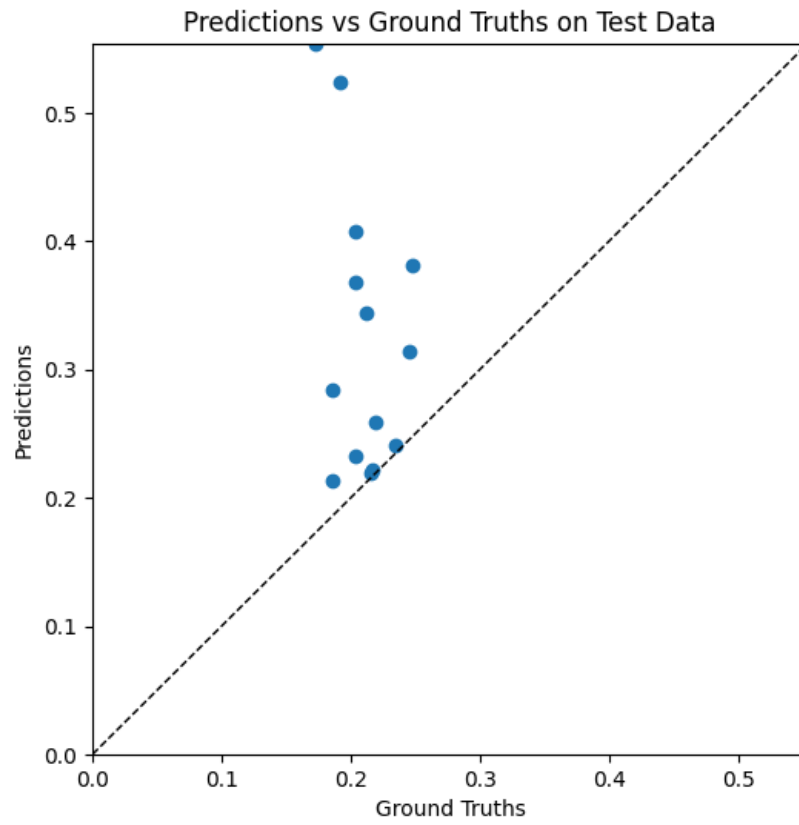


Fig. 4. The model's predictions plotted against ground truths for CHASE

5 Accomplishments

Using annotated fundus image datasets, we were able to pipeline the images into usable data for our models, creating a relatively accurate CNN model that predicted the disagreement value between annotators with low error and loss scores. However, we were only able to achieve this for the RIGA dataset, as the CHASE dataset did not have sufficient data points to properly train the model.

6 Contributions

- **Fu Jen Chiang:** I wrote the requirements, README.md and helped implement the function to save results in a folder. Contributed to the regression model by changing the loss function. Wrote the abstract, application, problem description and motivation. I created the slide and script for the demo video and edited it.
- **Ethan Saw:** I created code to parse the chase db1 dataset into image and disagreement values. This was later extended to put these results into a .csv used for model training but we figured that this dataset was unfortunately too small, so this code was adapted by Taewoong to parse the RIGA Bin-Rushed dataset instead, which we used for this project. I then added code to show the number of annotators for different datasets, added some logging code in our regression model, and fixed some incorrect paths. After looking at our results, I noticed that our predictions vs ground truths was not good, so I made changes to the parameters and adjusted the layers in our model in order to make our model's predictive power better. I also have reviewed numerous PR's for this semester too.
- **Truman Ho:** I coded a base CNN regression model using a pretrained resnet18 model by PyTorch. I implemented the training/validation and testing routines to start evaluating the model on our data. After some adjustments from group members on the model, I further improved it by adding batch normalization and dropout layers into the fully connected layer of the model and changed the optimizer to SGD from Adam. This stabilized the model, achieving an average MAE of 0.00944 over five training runs with no seed. I wrote code to generate visualizations of validation results, the Prediction VS Ground Truths test results and the training and validation loss curves. I extended Taewoong's code for best and worst predictions to show annotation overlays.
- **Max Liu:** I wrote code to parse the chase db dataset into binary and disagreement values. However, this code had some issues so it was not merged to main. I discussed with Ethan possible reasons for our poor predictions vs ground truths alongside other model changes. Finally, I worked on all sections of the report and did heavy editing and writing for sections 1, 2, 4, and 7 alongside performing my own independent research on papers covering fundus image segmentation.

- **Taewoong Choi:** I borrowed Ethan’s code to calculate disagreement values to calculate disagreement values for the BinRushed data set. I extended Truman’s regression model to also calculate various performance measures like the MAE, MSE, and correlation coefficient. Using these measures, I found the best and worst predictions and their respective images and saved them in a png. Worked on/edited sections 2,3,4,6,7,8 on the report.

7 Conclusion and Discussions

This report explored the use of CNNs to predict annotation disagreement in fundus images. By utilizing fundus image data sets with annotations [4][5], we were able to calculate the disagreement value between annotators which was then used to train and validate a CNN model for predicting disagreement. The resulting model trained on RIGA produced relatively good results with $MAE = 0.008137$, $MSE = 0.000129$, and $r = 0.81818$. However, because the disagreement values are initially small ($\bar{x} = 0.025$), MAE and MSE will subsequently be small as well. Looking at the correlation coefficient gives some context and tells us that our model’s predictions are not as close to ground truth as MAE and MSE may suggest. The model trained on CHASE was unusable due to a lack of data as mentioned above.

Although a correlation coefficient of 0.81818 is generally considered a moderate-high level of correlation, a more accurate model seems to be attainable when contextualized with RIGA’s ground truth.

8 Future Work

Using CNNs to filter out images with high levels of disagreement to clean and filter training data sets of other machine learning models has some promise. The model presented in this report was trained on a small data set and with the raw images. For future work, using larger data sets and using pre-processing techniques such as principal component analysis may lead to a more accurate model. Playing around with batch sizes, the amount of images the model is presented per epoch, and the number of epochs may also lead to a more accurate model. Otherwise, using a different CNN model with more layers such as resnet 50 may have a great impact on model accuracy.

Acknowledgments

This report/project was completed with the help of Professor Ghassan Hamarneh for guidance, feedback, and suggesting we use performance measures to quantify our model.

Appendix

1 BinRushed Data Acquisition

This appendix includes instructions for downloading the RIGA BinRushed data set.

To download, go to the website cited in [4] and scroll down to 'files' and click 'BinRushed.zip'. You will see that the data is split into 4 files BinRushed 1,2,3, and 4. This report uses data from BinRushed 2,3, and 4 only. The training data consists of the images from BinRushed 2 and 3, while the validation data is the images from BinRushed 4. The original images are labeled prime and the annotated images are labeled with '-1', '-2', etc after the image name.

2 CHASE Data Acquisition

This appendix includes instructions for downloading the CHASE data set.

To download, go to the website cited in [5] and scroll down to 'downloads' and download the Archive file labeled 'CHASEDB1.zip'.

References

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